

Conformer-CNN Heartbeat Classification on MIT-BIH with AAMI Evaluation and Calibration

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Abstract

We release an end-to-end ECG heartbeat classifier that converts beat-centered recordings from the MIT-BIH Arrhythmia Database into Morlet scalograms and applies a compact Conformer-CNN. The public pipeline reproduces training, evaluation, calibration, and robustness analyses using stored ‘Research_Runs’ artifacts. Hold-out accuracy reaches 0.60 (macro-F1 0.26), and all tables/figures are generated directly from scripts.

Keywords: ECG; Arrhythmia; Deep learning; Calibration; Reproducibility

1. Introduction

Wearable arrhythmia monitoring requires reproducible algorithms that report calibrated probabilities and expose robustness to sensor drift. We target the MIT-BIH Arrhythmia Database [1] under AAMI EC57 constraints, extending prior CNN/LSTM approaches [2] and attention systems [3] with auditable tooling.

2. Methods

The pipeline performs patient-level splits (38 train/validation, 7 test) as detailed in ??, maps annotations to five AAMI EC57 classes, computes streaming normalization, and renders 32-scale Morlet scalograms. A single Conformer block (128-d embeddings, eight heads) sits atop convolutional feature extractors, training for 20 epochs with class weighting. Evaluation scripts compute leaderboard metrics (??), per-class precision/recall (??), and calibration plus robustness summaries (??, ??).

3. Results

Compared to CNN-LSTM and attention-only baselines, the Conformer-CNN raises macro-F1 by at least 0.11 and preserves ventricular F1 at 0.57 (??). Confusion and PR plots (??) reveal SVEB and Fusion misclassifications, while calibration scores show conservative probabilities for minority classes. Gaussian noise perturbations degrade accuracy by only five percentage points at $\sigma=0.15$ (??).

4. Discussion

Minority-class weaknesses motivate targeted augmentation and potentially semi-supervised fine-tuning. Nevertheless, the scripted workflow—from TFRecord generation to LaTeX tables—offers a transparent baseline for community benchmarking and facilitates integration into telemetry systems that demand reliability evidence.

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5. Conclusion

We provide scripts, metrics, and figures for a Conformer-CNN heartbeat classifier emphasizing reproducibility, calibration, and robustness. Future work will extend to multi-lead Holter data and reliability-aware decision rules.

References

- [1] George B Moody and Roger G Mark. The impact of the mit-bih arrhythmia database. *IEEE Eng. in Med. and Biol.*, 20(3):45–50, 2001.
- [2] U R Acharya et al. A deep convolutional neural network model to classify heartbeats. *Computers in Biology and Medicine*, 89:389–396, 2017.
- [3] P Rajpurkar et al. Cardiologist-level arrhythmia detection with deep neural networks. *arXiv preprint arXiv:1707.01836*, 2017.